

$$\therefore U_0 = \frac{1}{RC} \int U_I dt = 10^3 \int U_I dt$$

$$\begin{aligned} \text{(d)} \quad U_0 &= -\frac{1}{C} \int \left( \frac{U_{I1}}{R_1} + \frac{U_{I2}}{R_2} \right) dt \\ &= -100 \int (U_{I1} + 0.5 U_{I2}) dt \end{aligned}$$

6.14. (1) 对  $A_1$ , 由电阻对称性可知

$$U_{O1} = \frac{R_4}{R_1} (U_0 - U_{I1}) = U_0 - U_{I1}$$

对  $A_2$ , 由“虚短”和“虚断”

$$\frac{U_{O1} - U_+}{R} = i_c = C \frac{du_c}{dt} = C \frac{du_-}{dt}$$

$$\because U_+ = U_- = U_0$$

$$\therefore \frac{-U_{I1}}{R} = C \frac{dU_0}{dt}$$

$$\therefore U_0 = -\frac{1}{RC} \int U_I dt = -10 \int U_I dt$$

$$(2) \because U_0 = -10 \int U_I dt = 10t + U_0(0) = 10t$$

$$\therefore U_0 = 6V \text{ 所需时间 } t = 0.6s$$

6.15 对  $A_2$ , 输出端电压为

$$U_{O2} = \left( 1 + \frac{R_2}{R_3} \right) U_0 = 2U_0$$

$$\text{对 } A_1, \quad U_{O2} = -\frac{1}{RC} \int U_I dt = -2 \int U_I dt$$

$$\therefore U_0 = -\int U_I dt$$

6.16 (1) 由“虚短”, “虚断”